

From Quantum Mechanics to Electromagnetism

Gauge covariance, Wilson lines, curvature, and Maxwell's equations

Student packet

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How to use this packet

This is not an introductory reading. It is a guided derivation for students who already know the Schrödinger equation, vector calculus, and enough linear algebra to tolerate index notation.

Do not read this passively. The problems are the packet. Keep a scratch notebook open, check the signs, and do not trust a formula until you know which convention produced it.

The goal is not merely to repackaging Maxwell's equations in fancier notation. The goal is to show how much of electromagnetism is forced on us once we take one quantum-mechanical claim seriously: the phase of a wave function is not directly observable.

That sounds innocent. In ordinary quantum mechanics we are told, repeatedly, that multiplying a wave function by a constant phase does not change the physics. Fine. But if that is really the principle, a reasonable student should immediately ask the more annoying question: why only a constant phase? Why not a phase that is different at every point in space and time?

The answer is that the probability density does not care, but the Schrödinger equation does. Derivatives notice when the phase varies. The whole packet is a guided attempt to repair that problem without cheating. The surprise is that the repair is not a small technical footnote; it is the beginning of electromagnetic gauge theory.

The derivational arc is:

1. Global phase is physically irrelevant.
2. Local phase changes destroy covariance of ordinary derivatives.
3. Repairing covariance forces us to introduce a new field A_i .
4. That new field is gauge-dependent, so it is not directly observable point-by-point.
5. Wilson lines explain how A can still have physical consequences.
6. The commutator of covariant momenta measures curvature.
7. In three-dimensional language, that curvature is the magnetic field.
8. The Aharonov–Bohm effect explains why this is not just formal symbolism.
9. Spacetime covariance combines the scalar and vector potentials into a 1-form A .
10. The field strength $F = dA$ packages the electric and magnetic fields.
11. The identity $dF = 0$ gives the two source-free Maxwell equations.
12. A spacetime metric and a four-current give the sourced Maxwell equations.
13. Charge conservation follows from antisymmetry.

A convention warning: this packet first uses a dimensionless gauge potential. This is deliberate. The physical electromagnetic vector potential can be restored later by inserting the appropriate factor of charge, $\frac{q}{\hbar c}$, depending on the unit system. For now, keeping A dimensionless lets us see the geometry without stopping every three lines to argue about SI versus Gaussian units.

One more warning: some of the early moves will look ad hoc. That is not a defect. A large fraction of theoretical physics consists of making a suspicious-looking repair, checking whether it actually works, and then discovering that the repair was trying to tell you something deeper all along.

0.1 Global phase is not enough

In quantum mechanics, we are repeatedly told that the overall phase of the wave function is meaningless. The usual argument is that the physical interpretation of the wave function is not in the complex number Ψ itself, but in its Hermitian square.

For example, in the position-space representation, the state is described by a complex function $\Psi(t, \mathbf{x})$, and the probability density is

$$|\Psi(t, \mathbf{x})|^2.$$

If we rotate the phase by a real constant α ,

$$\Psi(t, \mathbf{x}) \mapsto \Psi'(t, \mathbf{x}) = e^{i\alpha} \Psi(t, \mathbf{x}),$$

then the probability density is unchanged. So the descriptions in terms of Ψ and $e^{i\alpha} \Psi$ are physically indistinguishable.

This is all fine, but it should bother you a little. If the probability-density argument is the whole story, then why not take

$$\Psi'(t, \mathbf{x}) = e^{i\alpha(t, \mathbf{x})} \Psi(t, \mathbf{x})$$

where α is an entire function rather than one number? The probability density is still identical point-by-point. The catch is that the Schrödinger equation contains derivatives, and derivatives are exactly the operation that notices when something varies from point to point.

This is the first serious lesson of gauge theory: the quantity we want to call physically irrelevant can be made local, but only if the equations of motion are rebuilt so that every term varies in the same way. That common behavior is what we mean here by covariance.

Problem 1: Global phase and probability density

This first check is supposed to be easy, but it is not supposed to be meaningless.

Compute $|\Psi'|^2$ when $\Psi' = e^{i\alpha} \Psi$ with constant real α . Then repeat the reasoning when $\alpha = \alpha(t, \mathbf{x})$ is a real function. The point is to isolate exactly what the probability-density argument does and does not prove.

Problem 2: Ordinary derivatives are not covariant

This is where the problem first appears.

Assume

$$\Psi' = e^{i\alpha(\mathbf{x})} \Psi,$$

where α is time-independent but spatially varying. Compute $\partial_i \Psi'$ carefully enough that the product-rule term cannot hide. Then apply

$$\hat{p}_i = -i\hbar\partial_i$$

to Ψ' and identify the term that prevents $\hat{p}_i\Psi$ from transforming in the same way as Ψ .

0.2 Covariant derivatives and the gauge potential

For the moment, we simplify the problem by allowing α to depend on position but not on time. Then the time derivative in the Schrödinger equation is not the immediate problem. The spatial momentum operator is.

The ordinary momentum operator

$$\hat{p}_i = -i\hbar\partial_i$$

does not transform nicely under local rephasing because $\partial_i\alpha \neq 0$ gets caught by the derivative. The operator $\hat{p}_i$ is not doing anything wrong; it is applying the product rule. The phase is what changed, and the derivative noticed.

Can we fix this somehow? We introduce a new derivative

$$D_i = \partial_i - iA_i,$$

where $A_i(\mathbf{x})$ is a new field. At this stage A_i looks like a mischievous piece of bookkeeping. It is not yet obviously a physical field. The demand is simple: $D_i\Psi$ should transform the same way Ψ transforms:

$$D'_i\Psi' = e^{i\alpha}D_i\Psi.$$

This demand determines how A_i must transform. We are not guessing a new law of nature yet; we are asking what has to be true if local rephasing is to be a harmless redundancy in our description.

After that, we convert the repaired derivative into a repaired momentum,

$$\hat{\pi}_i = -i\hbar D_i.$$

In the original rough notebook this was jokingly called a delinquent momentum. The mature name is covariant momentum, but the joke is useful: ordinary momentum got into trouble with a local phase, and $\hat{\pi}_i$ is the replacement that knows how to stay out of trouble.

Problem 3: Derive the gauge transformation of A_i

Do not guess the transformation rule for A_i ; derive it from the demand that the new derivative repair the failure you just found.

Let

$$D_i = \partial_i - iA_i, \quad D'_i = \partial_i - iA'_i.$$

Assume $\Psi' = e^{i\alpha}\Psi$. Determine the transformation law for A'_i that makes

$$D'_i \Psi' = e^{i\alpha} D_i \Psi$$

hold for arbitrary Ψ .

Problem 4: Covariant momentum

Translate the repaired derivative into the corresponding repaired momentum.

Define

$$\hat{\pi}_i = -i\hbar D_i.$$

First express $\hat{\pi}_i$ in terms of $\hat{p}_i$ and A_i . Then use the covariance of $D_i \Psi$ to determine how $\hat{\pi}_i \Psi$ transforms. The goal is to check that this operator really deserves to replace ordinary momentum in the Schrödinger equation.

Problem 5: A gauge-covariant Hamiltonian

The Hamiltonian is only repaired if every term in the Schrödinger equation transforms together.

Suppose

$$\hat{H} = \frac{1}{2m} \hat{\pi}^2 + V(\mathbf{x}),$$

with V gauge-invariant. Explain why applying two covariant momenta in succession still gives an object that transforms like Ψ , and use that to show that $\hat{H}\Psi$ is covariant.

So the stupid-looking idea worked. We introduced a new field A_i , forced it to transform in a strange non-covariant way, and in exchange the equation became covariant. That is suspicious enough to deserve a closer look.

0.3 Wilson lines and gauge-invariant wave functions

Now that we have introduced A_i , we should try to understand what we actually did. We began with a wave function Ψ whose phase is ambiguous:

$$\Psi \sim e^{i\alpha} \Psi.$$

Can we simply dress the wave function in a way that gets rid of this ambiguity? Surprisingly, yes, provided we compare phases by integrating the gauge potential along a path.

Let γ be a path from a reference point q to a point p . Consider the line integral

$$\int_{\gamma} A = \int_q^p A_i dx^i.$$

Under the gauge transformation $A'_i = A_i + \partial_i \alpha$, the fundamental theorem of line integrals gives

$$\int_q^p A' = \int_q^p A + \alpha(p) - \alpha(q).$$

This is exactly the kind of endpoint phase shift we need. If we choose the reference point so that $\alpha(q) = 0$, then the Wilson-line factor

$$W(q, p; \gamma) = \exp \left(-i \int_q^p A \right)$$

cancels the local phase of $\Psi(p)$. The price is that the dressed wave function depends on the path γ . That path-dependence is not an annoyance to sweep under the rug; it is the first hint of curvature.

Problem 6: Wilson-line transformation

The Wilson line is designed to remember how phase changes along a path.

For

$$W(q, p; \gamma) = \exp \left(-i \int_q^p A \right),$$

use $A'_i = A_i + \partial_i \alpha$ and the fundamental theorem of line integrals to compute $W'(q, p; \gamma)$. Keep both endpoint terms $\alpha(p)$ and $\alpha(q)$; dropping one too early hides the mechanism.

Problem 7: A gauge-invariant dressed wave function

Now use the Wilson line to remove the local phase ambiguity at the endpoint.

Assume the reference point q is chosen so that $\alpha(q) = 0$ and define

$$\widetilde{\Psi}(p) = W(q, p; \gamma) \Psi(p).$$

Using the result of the previous problem, verify that $\widetilde{\Psi}(p)$ is gauge-invariant. Then state clearly what new dependence has been introduced by this construction.

For a closed loop, the reference point and endpoint are the same. The phase factor becomes

$$W(\gamma) = \exp \left(-i \oint_{\gamma} A \right).$$

By Stokes' theorem, this can be written as a surface integral of the curvature:

$$\oint_{\gamma} A = \int_{\Sigma} F, \quad \partial \Sigma = \gamma.$$

This is the mathematical skeleton of the Aharonov–Bohm effect.

0.4 The Aharonov–Bohm experiment

The Aharonov–Bohm experiment is the moment where this stops looking like notation and starts looking like physics.

Imagine a double-slit-style experiment, but with a long solenoid placed between the two electron paths. The magnetic field can be confined to the inside of the solenoid, so the electrons travel through a region where the local magnetic field is essentially zero. Classically, one might expect no magnetic influence on the electrons along the paths.

But the wave function does not merely carry local force information. It carries phase. As the electron travels, its phase is dressed by a Wilson line. When the two paths are recombined, the relative phase is controlled by the closed-loop quantity

$$\Delta\varphi = - \oint_{\gamma} A.$$

By Stokes' theorem,

$$\oint_{\gamma} A = \int_{\Sigma} F, \quad \partial\Sigma = \gamma.$$

In physical units this is proportional to the magnetic flux through the solenoid. So the interference pattern can shift even though the electrons never pass through a region with nonzero local magnetic field.

This is the weird but essential point: the value of A_i at a single point is not directly observable, because it can be changed by a gauge transformation. Nevertheless, the closed-loop phase can be observable. The vector potential is less real than the magnetic field in one sense, because it is gauge-dependent; but it is more fundamental in another sense, because quantum particles couple to the potential itself.

Problem 8: Interpreting the Aharonov–Bohm phase

The tempting but wrong conclusion is that the Aharonov–Bohm effect proves the value of A_i at a point is directly measurable.

Explain why that cannot be right. Then identify the gauge-invariant quantity that the interference experiment actually measures, and explain why it can be nontrivial even when the local magnetic field vanishes along the electron paths.

0.5 Curvature and the magnetic field

We have now met a gauge potential A_i and a Wilson-line phase built from it. The next question is what part of A_i is gauge-invariant locally.

The answer is its curvature:

$$F_{ij} = \partial_i A_j - \partial_j A_i.$$

This is the same antisymmetric combination that appears in the ordinary curl. In fact, in three-dimensional Euclidean language this curvature is just another way of writing the magnetic field. But before we translate it into vector calculus, we should see where it appears naturally in quantum mechanics.

It appears in the failure of covariant derivatives to commute.

That is a major conceptual point. Ordinary partial derivatives commute, so $[\partial_i, \partial_j] = 0$. Covariant derivatives generally do not. The commutator is not a nuisance term; it measures curvature.

Problem 9: Commutator of covariant derivatives

This calculation is the local version of the Wilson-loop story.

Using

$$D_i = \partial_i - iA_i,$$

compute $D_i D_j \Psi - D_j D_i \Psi$ without skipping the product-rule terms. Show that all terms except the antisymmetric derivative of A cancel, leaving

$$[D_i, D_j] \Psi = -iF_{ij} \Psi.$$

Problem 10: Commutator of covariant momenta

Convert the curvature calculation from covariant derivatives to covariant momenta.

Using $\hat{\pi}_i = -i\hbar D_i$, determine $[\hat{\pi}_i, \hat{\pi}_j] \Psi$ in terms of F_{ij} . Track the factor of i and the powers of $\hbar$ explicitly.

In three-dimensional Euclidean space, the magnetic field is the vector dual to the spatial curvature:

$$B^k = \frac{1}{2} \varepsilon^{kij} F_{ij}.$$

Equivalently,

$$F_{ij} = \varepsilon_{ijk} B^k.$$

Thus

$$\mathbf{B} = \nabla \times \mathbf{A}.$$

Wait what? We started by worrying about an arbitrary local phase in a quantum wave function, and now we are staring at the magnetic field. This is exactly the sort of result that should make you suspicious in the productive way: either we made a sign error, or we have discovered that the usual vector potential was never just a computational convenience.

Problem 11: The magnetic field from curvature

This is where the antisymmetric tensor becomes the magnetic field students know from vector calculus.

Assume

$$F_{ij} = \partial_i A_j - \partial_j A_i, \quad B^k = \frac{1}{2} \varepsilon^{kij} F_{ij}.$$

Compute the three components B^1 , B^2 , and B^3 and compare them with the standard components of $\nabla \times \mathbf{A}$.

Problem 12: The first Maxwell equation

Use the fact that the magnetic field has appeared as a curl to derive a Maxwell equation.

Starting from

$$B^i = \varepsilon^{ijk} \partial_j A_k,$$

compute $\partial_i B^i$. Explain why the result vanishes because of the conflict between symmetric second derivatives and the antisymmetry of ε^{ijk} .

0.6 Why vector calculus is hiding the structure

A lot just happened. We started with the local phase of a quantum wave function and stumbled into something that looks exactly like the magnetic field. That is already strange. It is even stranger that we did it without beginning from classical electrodynamics at all.

At this point it is tempting to say, “Great, $\mathbf{B} = \nabla \times \mathbf{A}$, moving on.” But that would hide the real structure.

The cross product is one of the most misleadingly familiar operations in vector calculus. It is useful, but it is also a three-dimensional accident. What we really constructed was not fundamentally a vector. We constructed an antisymmetric object with two lower indices:

$$F_{ij} = \partial_i A_j - \partial_j A_i.$$

Equivalently, A_i is better understood as the component of a 1-form,

$$A = A_i dx^i,$$

and its curvature is the 2-form

$$F = dA = \frac{1}{2} F_{ij} dx^i \wedge dx^j.$$

In three-dimensional oriented Euclidean space, a 2-form has three independent components, so we can disguise it as a vector by using the Levi-Civita symbol. That disguise is the magnetic field vector. In other dimensions the disguise fails, but the 2-form survives.

This section is partly undoing some damage from multivariable calculus. Gradients, curls, dot products, and cross products are not all equally primitive. The differential df and the exterior derivative dA make sense before any metric is introduced. The gradient and cross product require extra geometric choices.

Problem 13: Levi-Civita contraction

The following identity is the algebraic engine behind converting between antisymmetric tensors and vectors in three dimensions:

$$\varepsilon^{ijk}\varepsilon_{klm} = \delta_\ell^i\delta_m^j - \delta_m^i\delta_\ell^j.$$

Prove it by considering the possible index coincidences and permutations. Then use it to check that the definitions

$$B^k = \frac{1}{2}\varepsilon^{k\ell m}F_{\ell m} \quad \text{and} \quad F_{ij} = \varepsilon_{ijk}B^k$$

are inverse to each other when F_{ij} is antisymmetric.

Problem 14: What is special about three dimensions?

The usual expression $\nabla \times \mathbf{A}$ is familiar, but familiarity is not the same as fundamentality.

Explain why the curl of a vector field is a three-dimensional repackaging of an antisymmetric two-index object. Then identify the object that still makes sense in two, four, or arbitrary dimensions.

0.7 The spacetime gauge potential

Earlier we restricted to time-independent rephasings because it let us focus on the spatial momentum operator. Now we have to pay that debt.

The easy way forward—with hindsight doing a lot of work—is to return to the Wilson-line dressing factor. In space we integrated

$$A_i dx^i.$$

If the phase can also vary in time, then the line integral should have a time component too. Since ct has the same physical dimension as x , y , and z , define

$$x^0 = ct, \quad \partial_0 = \frac{\partial}{\partial x^0} = \frac{1}{c} \frac{\partial}{\partial t}.$$

We combine the scalar and vector potentials into the spacetime 1-form

$$A = A_\mu dx^\mu = -\Phi dx^0 + A_i dx^i.$$

Thus

$$A_0 = -\Phi.$$

The sign is a convention, but it is not arbitrary once we decide how we want the electric field and Maxwell equations to come out. We are using the convention

$$A'_\mu = A_\mu + \partial_\mu \alpha.$$

The repeated-index notation here is Einstein summation convention: repeated upper/lower index pairs are summed. We will use this convention heavily from this point onward.

Problem 15: Scalar-potential transformation

The spatial transformation law $A'_i = A_i + \partial_i \alpha$ now has to be extended to the time component.

Using

$$A_0 = -\Phi, \quad A'_0 = A_0 + \partial_0 \alpha,$$

derive the transformation law for Φ . Be explicit about the conversion

$$\partial_0 = \frac{1}{c} \frac{\partial}{\partial t}.$$

Problem 16: Spacetime Wilson line

Repeat the Wilson-line argument in spacetime.

For

$$W(q, p; \gamma) = \exp \left(-i \int_\gamma A_\mu dx^\mu \right),$$

use $A' = A + d\alpha$ to determine how W transforms between endpoints q and p . Then explain why $W(q, p; \gamma)\Psi(p)$ is gauge-invariant when $\alpha(q) = 0$.

0.8 The Faraday-Maxwell tensor

Now we return to the question that broke the naive four-dimensional version of Green's theorem. In three dimensions we could write things like

$$\oint_\gamma \mathbf{A} \cdot d\mathbf{x} = \int_\Sigma (\nabla \times \mathbf{A}) \cdot \mathbf{n} da.$$

Looks nice. But in four dimensions the symbols become suspect. What is the vector normal to a surface in spacetime? What is a four-dimensional cross product? Are we claiming Stokes' theorem only works in three dimensions?

The problem is the vector-calculus disguise. The derivative naturally has an index down, so the four-dimensional replacement for the curl is the antisymmetric derivative

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

This object has many names: Maxwell tensor, Faraday tensor, Faraday-Maxwell tensor, electromagnetic field-strength tensor. The name matters less than the structure. It is the curvature of the spacetime gauge potential:

$$F = dA = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu.$$

The wedge product is doing the job that the cross product was pretending to do in three dimensions. It remembers orientation and antisymmetry without needing a special three-dimensional accident.

Problem 17: Gauge invariance of $F_{\mu\nu}$

The Faraday-Maxwell tensor is supposed to be the gauge-invariant curvature of A .

Starting from

$$A'_\mu = A_\mu + \partial_\mu \alpha,$$

compute

$$F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu$$

and show exactly where the extra terms cancel.

Problem 18: Components of $F_{\mu\nu}$

Now translate the spacetime tensor back into electric and magnetic fields.

Using

$$A_0 = -\Phi, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu,$$

compute F_{0i} and F_{ij} . Then define $E_i = -F_{0i}$ and verify

$$E_i = -\partial_i \Phi - \frac{1}{c} \frac{\partial A_i}{\partial t}, \quad F_{ij} = \varepsilon_{ijk} B^k.$$

With this convention,

$$(F_{\mu\nu}) = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}.$$

0.9 The source-free Maxwell equations

The source-free Maxwell equations are the equations that follow from the simple identity

$$dF = 0.$$

This is not a dynamical assumption yet. Since $F = dA$, it is really the statement

$$d^2A = 0.$$

In vector-calculus language, the same statement becomes two familiar laws:

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\frac{1}{c} \frac{\partial \mathbf{B}}{\partial t}.$$

The first says that magnetic field lines do not begin or end: there are no magnetic monopoles in ordinary Maxwell theory. The second is Faraday's law of induction. The point of this section is to see both laws as different faces of the same compact spacetime equation.

Problem 19: No magnetic monopoles

Extract the purely spatial part of $dF = 0$.

Take the cyclic identity

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0$$

with all three indices spatial, contract with the Levi-Civita symbol, and show that it is equivalent to

$$\nabla \cdot \mathbf{B} = 0.$$

Problem 20: Faraday's law

Now extract the mixed time-space part of the same identity.

Set one index equal to 0 and the other two spatial. Use

$$F_{0i} = -E_i, \quad F_{ij} = \varepsilon_{ijk} B^k,$$

to derive

$$\nabla \times \mathbf{E} = -\frac{1}{c} \frac{\partial \mathbf{B}}{\partial t}.$$

Then interpret the integral form as Faraday's law of induction.

0.10 Exterior derivatives and Stokes' theorem

We have now used exterior forms enough that it is worth pausing to name the machinery.

A differential exterior form is the natural object to integrate over an oriented geometric region. A 0-form is a function. A 1-form integrates over a curve. A 2-form integrates over a surface. A 3-form integrates over a volume. The degree of the form tells you the dimension of the thing it naturally integrates over.

The exterior derivative d raises degree by one:

$$\omega \in \Omega^p \implies d\omega \in \Omega^{p+1}.$$

The general Stokes theorem is

$$\int_U d\omega = \int_{\partial U} \omega.$$

This single theorem includes the fundamental theorem of calculus, Green's theorem, the Kelvin-Stokes theorem, and the divergence theorem as special cases. The multivariable-calculus theorems were never separate miracles; they were shadows of one statement.

Problem 21: Why $d^2 = 0$

Prove the formal identity that has been doing most of the work.

Let ω be a p -form. Compute $d(d\omega)$ in components and explain why the expression vanishes: the second partial derivatives are symmetric, while the wedge product is antisymmetric.

Then state the local consequence for a closed 2-form: on a contractible region, $dF = 0$ implies $F = dA$ for some 1-form A .

Problem 22: The boundary of a boundary

Use Stokes' theorem and $d^2 = 0$ to prove the slogan:

the boundary of a boundary is empty.

More precisely, argue that $\partial(\partial U)$ must be zero in the sense detected by integrating arbitrary forms over it.

0.11 Charges, currents, and the sourced Maxwell equations

So far we have derived the two Maxwell equations that come from $F = dA$. These are the source-free equations. To get the other two equations, we need charges and currents.

Gauss's law says that the electric flux through a closed surface measures the charge inside:

$$Q(\Sigma) = \int_{\Sigma} \mathbf{E} \cdot d\mathbf{a}.$$

If $\Sigma = \partial U$, then the divergence theorem rewrites this as

$$Q(U) = \int_U \nabla \cdot \mathbf{E} \, dv.$$

If charge is distributed continuously, we write

$$dQ = \rho(\mathbf{x}) \, dx \wedge dy \wedge dz,$$

so that

$$Q(U) = \int_U \rho \, dx \wedge dy \wedge dz.$$

The goal is to recognize Gauss's law and the Ampère-Maxwell law as components of one spacetime equation. With the sign conventions used in this packet, the clean form is

$$\partial_{\mu} F^{\nu\mu} = j^{\nu}.$$

This is equivalent to $\partial_{\mu} F^{\mu\nu} = -j^{\nu}$. The difference is not physics; it is bookkeeping forced by the earlier sign conventions.

Problem 23: Gauss's law

Start with the integral form of Gauss's law and recover the local equation.

If

$$Q(U) = \int_U \rho d^3x \quad \text{and} \quad Q(U) = \int_U \nabla \cdot \mathbf{E} d^3x$$

for every reasonable region U , explain why the integrands must agree. Then identify the temporal component of

$$\partial_\mu F^{\nu\mu} = j^\nu$$

that reproduces this result.

Problem 24: The Ampère-Maxwell law

Write out the spatial components $\nu = i$ of

$$\partial_\mu F^{\nu\mu} = j^\nu.$$

Use the conventions

$$F^{i0} = -E_i, \quad F^{ij} = F_{ij} = \varepsilon_{ijk} B^k, \quad j^i = \frac{1}{c} j_{\text{phys}}^i,$$

and show that the result is

$$c \nabla \times \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t} + \mathbf{j}_{\text{phys}}.$$

Then explain which term is Maxwell's displacement-current correction.

0.12 The spacetime metric and raising indices

The sourced equation used $F^{\nu\mu}$, but our field strength was originally $F_{\mu\nu}$. Moving indices is not decorative notation. It requires a metric.

In ordinary Euclidean 3-space, the metric is so invisible that students often do not realize it is being used. In spacetime it becomes impossible to hide. We need a bilinear form $\eta_{\mu\nu}$ and its inverse $\eta^{\mu\nu}$ satisfying

$$\eta^{\mu\lambda} \eta_{\lambda\nu} = \delta_\nu^\mu.$$

Then

$$F^{\mu\nu} = \eta^{\mu\rho} \eta^{\nu\sigma} F_{\rho\sigma}.$$

If $F^{\mu\nu}$ were independent from $F_{\mu\nu}$, then we would effectively have two electromagnetic fields. In vacuum we do not. So the raising operation must relate them. The signs in that relation force the mostly-plus Minkowski metric:

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \text{diag}(-1, 1, 1, 1).$$

In materials, this story becomes more complicated and is described by constitutive relations. In vacuum, the metric is the relation.

Problem 25: Deriving the mostly-plus metric

This problem is about signs, not about memorizing a metric.

Assume the spatial metric is Euclidean and that raising both indices on $F_{\mu\nu}$ should give the $F^{\mu\nu}$ used in the sourced Maxwell equation. Compare the signs of the electric and magnetic components before and after raising indices. Use this comparison to determine why the convention already chosen forces

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \text{diag}(-1, 1, 1, 1).$$

0.13 The Hodge star

Once a metric and an orientation are available, there is one more important operation: the Hodge star.

The Hodge star takes a p -form in n dimensions to an $(n - p)$ -form. It uses the metric to raise indices and the volume form to trade the remaining indices for their complementary set. In four-dimensional spacetime, it maps 2-forms to 2-forms, so it can act directly on the Faraday-Maxwell tensor.

For a p -form ω , the component definition is

$$(*\omega)_{i_1 \dots i_{n-p}} = \frac{1}{p!} \varepsilon_{i_1 \dots i_{n-p}}{}^{j_1 \dots j_p} \omega_{j_1 \dots j_p}.$$

For electromagnetism, the Hodge star is the operation that exchanges the electric and magnetic parts of the field, with signs determined by the Lorentzian metric. This is also where the metric-dependent half of Maxwell's equations naturally lives.

Problem 26: Raising the spacetime volume form

The sign of the spacetime volume form changes when all indices are raised with the mostly-plus metric.

Assume

$$\varepsilon_{0123} = +1.$$

Compute ε^{0123} and explain which metric component is responsible for the sign.

Problem 27: The Hodge dual of F

Compute how the Hodge star acts on the electromagnetic field-strength tensor.

Using

$$(*F)_{\mu\nu} = \frac{1}{2} \varepsilon_{\mu\nu}{}^{\rho\sigma} F_{\rho\sigma},$$

show that

$$(*F)_{0i} = B_i, \quad (*F)_{ij} = \varepsilon_{ijk} E^k.$$

Then describe in words what $*$ does to the electric and magnetic parts of F .

Problem 28: Applying the Hodge star twice

Apply the Hodge star twice to a 2-form in four-dimensional Minkowski spacetime.

Use the result of the previous problem, or compute directly from the Levi-Civita contractions, to determine the sign in

$$**F = ?$$

Then compare this with the Euclidean intuition that a ninety-degree rotation twice might give a minus sign.

In differential-form notation, the vacuum Maxwell equations can be written compactly as

$$dF = 0, \quad d*F = J,$$

where J is the appropriate current 3-form. This form makes clear that the first equation is metric-independent while the second uses the Hodge star and therefore the metric.

0.14 Charge conservation

The original notebook ended here with a note to finish the argument. This is actually the right place to end, because the final result is one of the strongest consistency checks in the whole packet.

Maxwell's equations should not allow charge to disappear. Locally, charge conservation means that any decrease of charge density in a region must be accounted for by current flowing out of that region. In spacetime notation this is the continuity equation

$$\partial_\nu j^\nu = 0.$$

The remarkable point is that this is not an additional Maxwell equation. It follows from the sourced Maxwell equation and the antisymmetry of $F^{\nu\mu}$.

Problem 29: Local charge conservation

Derive local charge conservation from the sourced Maxwell equation.

Start with

$$\partial_\mu F^{\nu\mu} = j^\nu.$$

Take ∂_ν of both sides and use the antisymmetry of $F^{\nu\mu}$ to show

$$\partial_\nu j^\nu = 0.$$

Finally rewrite this in terms of ρ and $\mathbf{j}_{\text{phys}}$.

Problem 30: Conservation of total charge

Turn the local continuity equation into a global conservation law.

Assume ρ and $\mathbf{j}_{\text{phys}}$ fall off fast enough at spatial infinity, or vanish outside a sufficiently large ball. Define

$$Q(t) = \int_{\mathbb{R}^3} \rho(t, \mathbf{x}) d^3x.$$

Use the divergence theorem to show that $dQ/dt = 0$.

0.15 Synthesis

Problem 31: What did local phase force us to introduce?

Write a short conceptual answer, not a calculation.

Starting from the demand that local phase redefinitions should not change the physics, list the new mathematical structures that were forced into the theory: the gauge potential, Wilson lines, curvature, and the electromagnetic field.

Problem 32: Geometry versus metric

Separate the two kinds of structure that appeared in this packet.

Which parts of Maxwell theory came from exterior calculus and gauge curvature before any metric was used? Which parts required the spacetime metric, index-raising, or the Hodge star?

0.16 Further reading

Useful references for context:

- Gauge theory, for the general framework of local symmetry and gauge fields.
- Aharonov–Bohm effect, for the physical experiment motivating Wilson-loop phases.
- Electromagnetic tensor, for standard component conventions.
- Differential form, for the language of A , $F = dA$, and $dF = 0$.
- Hodge star operator, for the metric-dependent operation used to write the sourced Maxwell equations.

0.17 Convention summary

The packet used the following sign conventions:

$$\Psi' = e^{i\alpha}\Psi, \quad D_i = \partial_i - iA_i, \quad A'_i = A_i + \partial_i\alpha.$$

$$\hat{\pi}_i = -i\hbar D_i = \hat{p}_i - \hbar A_i.$$

$$A = A_\mu dx^\mu = -\Phi dx^0 + A_i dx^i, \quad x^0 = ct.$$

$$F = dA, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

$$E_i = -F_{0i}, \quad F_{ij} = \varepsilon_{ijk} B^k.$$

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \text{diag}(-1, 1, 1, 1).$$

$$\partial_\mu F^{\nu\mu} = j^\nu, \quad j^0 = \rho, \quad j^i = \frac{1}{c} j_{\text{phys}}^i.$$

With these conventions, Maxwell's equations are

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\frac{1}{c} \frac{\partial \mathbf{B}}{\partial t},$$

and

$$\nabla \cdot \mathbf{E} = \rho, \quad c \nabla \times \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t} + \mathbf{j}_{\text{phys}}.$$